Locafy: An Open-Source, Zero-Cost Geospatial Framework for Hyperlocal Small Business Discovery in Developing Digital Economies

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ABSTRACT— In emerging economies such as India, micro, small, and medium enterprises (MSMEs) face severe economic and technological barriers to digital visibility. Centralized directory and delivery conglomerates impose steep commissions and annual subscription paywalls, while proprietary cloud mapping APIs (e.g., Google Maps Platform, Places API) compound recurring operational metering costs that render community platforms unviable. This paper presents Locafy, a production-grade, open-source hyperlocal business discovery and lead attribution web system engineered specifically for Indian commercial ecosystems. Locafy eliminates all commercial API dependencies by integrating OpenStreetMap spatial raster tiles, Leaflet GIS client rendering, and browser-native W3C Geolocation. At its computational core, the platform introduces a dual-phase query processing engine: (1) an optimized spherical trigonometry Haversine distance calculator paired with bounded SQL spatial pruning, and (2) a real-time temporal operating availability evaluator operating deterministically within the Asia/Kolkata (IST, UTC+05:30) timezone. The backend implements a resilient database adapter that transparently leverages Node.js v22 native node:sqlite (DatabaseSync) with automated fallback to better-sqlite3, eliminating external native compilation toolchains. Evaluated over an authentic empirical dataset comprising 49 verified commercial stores across 13 major Indian metropolitan regions, the system achieves sub-15ms query execution times, a 73.9% payload reduction via static asset decoupling, and 100% deterministic pass rates across end-to-end multi-actor verification cycles.

Keywords *— Hyperlocal Discovery, Geospatial Information Systems (GIS), Haversine Formula, Zero-Cost Architecture, OpenStreetMap, Leaflet.js, Lead Attribution, SQLite, Indian MSMEs.*

1. INTRODUCTION & BACKGROUND

India’s domestic retail and service economy comprises over 63 million micro, small, and medium enterprises (MSMEs), representing the primary livelihood source for over 110 million citizens and contributing upwards of 30% to national GDP. While rapid digital penetration via the Unified Payments Interface (UPI) and subsidized 4G/5G mobile connectivity has integrated consumers into the digital financial ecosystem, consumer discovery of neighborhood businesses remains severely centralized.

Commercial directory conglomerates and food-delivery aggregators exploit this asymmetry by imposing steep onboarding charges, per-lead paywalls, and high commission rates (18% to 30%). Simultaneously, independent developers, municipal organizations, or college project initiatives seeking to build open, community-oriented local directories are hindered by compounding cloud API metering tariffs. For example, the Google Maps Platform levies $5.00 to $7.00 per 1,000 Dynamic Map loads and $17.00 per 1,000 Nearby Search requests.

To resolve this challenge, this paper introduces Locafy, a zero-cost, open-architecture local business discovery and lead attribution platform designed specifically for the unique geographical and commercial landscape of India. Locafy operates end-to-end without paid external API keys, commercial database hosts, or heavyweight JavaScript frameworks.

2. LITERATURE REVIEW & THEORETICAL FOUNDATIONS

Geographic Information Systems (GIS) for local business search historically rely on client-server spatial querying over spatial indexes such as R-Trees or Geohashes. While modern spatial databases like PostgreSQL with PostGIS offer sophisticated planar and spheroidal operations, they demand dedicated database server daemons with non-trivial memory overhead.

In lightweight or embedded server contexts, SQLite provides an ideal zero-configuration, single-file relational engine. By formulating spherical geodesic equations directly within standard SQL and procedural runtime layers, high-throughput spatial querying is achievable without specialized spatial extensions.

3. SYSTEM ARCHITECTURE & ENGINEERING DESIGN

The architectural design of Locafy follows a tiered client-server model engineered for minimal latency and zero third-party dependencies. It comprises four decoupled layers:

- Client Presentation Layer: Pure Vanilla ES6 JavaScript modules with responsive CSS Custom Properties tokens, Leaflet.js map synchronization, and live W3C Geolocation detection.
- Gateway & Security Layer: Express.js server on port 3000, Helmet security headers, CORS protection, express-rate-limit throttling, and HMAC-SHA256 JWT role-based access control (RBAC).
- Core Service Controller Layer: Spatial Haversine search, real-time IST temporal status evaluation, lead attribution

interceptors, and 30-day aggregated merchant analytics.

- Persistence & Storage Layer: Resilient SQLite database adapter (node:sqlite / better-sqlite3) operating directly on locafy.db (156 KB) with parameterized SQL queries.

3.1 Resilient Database Driver Abstraction

Standard Node.js deployments on Windows or minimal Linux environments frequently fail when compiling native SQLite bindings (such as better-sqlite3) due to missing Python, Visual Studio C++ Build Tools, or node-gyp dependencies. Locafy resolves this vulnerability by implementing an automated driver detection adapter: if better-sqlite3 is compiled, it is utilized synchronously; otherwise, the runtime seamlessly falls back to the native Node.js v22 standard library node:sqlite module (DatabaseSync). The wrapper normalizes argument spreading, PRAGMA journal configurations, and statement execution semantics, guaranteeing 100% turnkey startup on any modern Node.js environment without external compilation steps.

4. MATHEMATICAL & ALGORITHMIC FORMULATIONS

4.1 Spherical Geodesic Distance via Haversine Formulation

Given a client's detected coordinate pair (phi_1, lambda_1) and a candidate business coordinate pair (phi_2, lambda_2), where phi denotes latitude and lambda denotes longitude in radians:

$$9C\text{Æ} = (lat2 - lat1) \cdot (< \text{ò } f \text{ 'Â } \text{";}^2 \text{Ò } \dagger\text{Æœs" Ò Æœs ' r } f\text{À} / 180) \quad (1)$$

The square of the half-chord length a between the two points is calculated as:

$$a = \sin^2(9C\text{Æ} / 2) + \cos(<c \text{ ' r } 6 \div 2 f\text{Æ}2) \cdot \sin^2(9C\text{»} / 2) \quad (2)$$

The angular distance in radians c and resultant surface distance d (using Earth's volumetric radius R = 6371.0 km) are:

$$c = 2 \cdot \text{atan2}(\text{" a, " } (1 - a)), \quad d = R \cdot c \ \ (km) \quad (3)$$

4.2 Real-Time Temporal Availability Engine

Operating hours are stored as structured JSON records containing daily shift intervals. Because local businesses frequently operate across midnight boundaries (e.g. dhabas or night clinics open from 18:00 to 03:00), simple scalar comparisons fail. Locafy evaluates temporal availability in four steps: (1) project system time into Asia/Kolkata timezone (UTC+05:30), (2) extract minute-of-day integer t_current = 60 · hour + minute in [0, 1439], (3) parse interval [t_open, t_close], and (4) evaluate boundary conditions:

- Standard Shift (t_open <= t_close): isOpen = (t_current >= t_open) AND (t_current <= t_close)
- Overnight Shift (t_open > t_close): isOpen = (t_current >= t_open) OR (t_current <= t_close)

4.3 Client-Side Intent Interception for Lead Attribution

Instead of routing customer phone calls or WhatsApp messages through expensive telephony proxy servers (such as Twilio or Exotel virtual numbers), Locafy implements Intent-Interception Proxies. When a consumer clicks Call Now, WhatsApp, or Directions, an asynchronous micro-request logs the lead type in the SQLite database with timestamp attribution before invoking the native mobile protocol handler (tel:, https://wa.me/, or Google Maps navigation).

5. DATABASE SCHEMA ARCHITECTURE

Table 1. Relational Schema Architecture and Spatial-Chronological Indexing of Locafy

Table Entity	Keys & Constraints	Key Attributes	Indexing Strategy
users	id (PK)	name, email (UK), phone, password_hash, role	Index on email
categories	id (PK)	name, slug (UK), icon, color	Index on slug
businesses	id (PK), owner_id (FK)	name, phone, whatsapp, lat, lng, is_verified, status	Compound (category, city, status, lat, lng)
reviews	id (PK), business_id (FK)	rating (1-5), comment, created_at	Compound UNIQUE (business_id, user_id)
leads	id (PK), business_id (FK)	type (view, call, whatsapp, directions, message)	Compound (business_id, created_at)
messages	id (PK), business_id (FK)	name, phone, email, body, status (new, read)	Index on (business_id, status)
favourites	user_id, business_id (PK)	created_at	Index on user_id

6. EMPIRICAL EVALUATION & EXPERIMENTAL RESULTS

The platform was seeded and tested with an authentic dataset encompassing 49 verified commercial stores, culinary

landmarks, and multi-speciality healthcare facilities distributed across 13 major Indian metropolitan regions (Delhi NCR, Mumbai, Bengaluru, Hyderabad, Kolkata, Chennai, Jaipur, Lucknow, Goa, Pune, Ahmedabad, Amritsar, and Chandigarh).

Static frontend profiling revealed that the original mockup embedded 86 KB raster images directly into HTML files as inline Base64 data strings. By decoupling assets into standalone PNG files (frontend/assets/logo.png) served with HTTP static caching headers, file payload dropped from 311 KB to 81 KB—representing a 73.9% size reduction.

Table 2. Empirical Performance Metrics Across Platform Subsystems

Evaluation Metric	Measured Value	Baseline Comparison	Operational Impact
Database Disk Footprint	156 Kilobytes	PostgreSQL (>50MB baseline)	Runs on micro instances (512MB RAM)
Static HTML Payload	81 Kilobytes	Original 311 KB (-73.9%)	Instant load on 3G/4G cellular networks
Search Query Latency	8.4 ms (Mean)	Cloud API (180-350 ms)	High throughput (>100 req/sec concurrent)
End-to-End Test Suite	4 / 4 Flows (100%)	Deterministic Pass	Verified Customer, Merchant, Admin & Analytics

7. SOCIETAL & ECONOMIC IMPACT ON INDIAN MSMES

In Tier-2, Tier-3, and rural Indian towns, independent service providers (such as electricians, motor mechanics, local bakeries, and clinics) lack the technical literacy and financial capital to maintain paid merchant accounts or buy online advertisement credits. By decoupling local discovery from proprietary cloud platforms, Locafy offers educational institutions, local trade bodies, and municipal governments a turnkey, free-software framework to host community discovery portals that operate permanently without recurring cloud bills.

8. FUTURE WORK & CONCLUSION

Future iterations of this research will explore: (1) integrating quantized local NLP vector embeddings via Transformers.js for semantic conversational querying (e.g. "quiet work cafe with fast wifi and vegan snacks"), (2) decentralized Sybil-resistant review verification using decentralized identifiers (DIDs), and (3) Progressive Web App (PWA) offline Service Worker caching for zero-connectivity catalog browsing in rural markets.

In conclusion, Locafy establishes that full-featured, responsive, and secure hyperlocal business directories with interactive mapping, live temporal evaluation, and lead attribution can be deployed with zero commercial API licensing overhead, demonstrating an accessible paradigm for decentralized digital commerce in developing nations.

9. REFERENCES

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